

17. $y'' - 4y = 2$; $y_1 = e^{-2x}$
 18. $y'' + y' = 1$; $y_1 = 1$
 19. $y'' - 3y' + 2y = 5e^{3x}$; $y_1 = e^x$
 20. $y'' - 4y' + 3y = x$; $y_1 = e^x$

Discussion Problems

21. (a) Give a convincing demonstration that the second-order equation $ay'' + by' + cy = 0$, a, b , and c constants, always possesses at least one solution of the form $y_1 = e^{m_1x}$, m_1 a constant.
 (b) Explain why the differential equation in part (a) must then have a second solution, either of the form $y_2 = e^{m_2x}$, or of the form $y_2 = xe^{m_1x}$, m_1 and m_2 constants.
 (c) Reexamine Problems 1–8. Can you explain why the statements in parts (a) and (b) above are not contradicted by the answers to Problems 3–5?

22. Verify that $y_1(x) = x$ is a solution of $xy'' - xy' + y = 0$. Use reduction of order to find a second solution $y_2(x)$ in the form of an infinite series. Conjecture an interval of definition for $y_2(x)$.

Computer Lab Assignment

23. (a) Verify that $y_1(x) = e^x$ is a solution of

$$xy'' - (x + 10)y' + 10y = 0.$$

- (b) Use (5) to find a second solution $y_2(x)$. Use a CAS to carry out the required integration.
 (c) Explain, using Corollary (a) of Theorem 3.1.2, why the second solution can be written compactly as

$$y_2(x) = \sum_{n=0}^{10} \frac{1}{n!} x^n.$$

3.3 Homogeneous Linear Equations with Constant Coefficients

Introduction We have seen that the linear first-order DE $y' + ay = 0$, where a is a constant, possesses the exponential solution $y = c_1 e^{-ax}$ on the interval $(-\infty, \infty)$. Therefore, it is natural to ask whether exponential solutions exist for homogeneous linear higher-order DEs

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \quad (1)$$

where the coefficients a_i , $i = 0, 1, \dots, n$ are real constants and $a_n \neq 0$. The surprising fact is that *all* solutions of these higher-order equations are either exponential functions or are constructed out of exponential functions.

Auxiliary Equation We begin by considering the special case of a second-order equation

$$ay'' + by' + cy = 0. \quad (2)$$

If we try a solution of the form $y = e^{mx}$, then after substituting $y' = me^{mx}$ and $y'' = m^2 e^{mx}$ equation (2) becomes

$$am^2 e^{mx} + bme^{mx} + ce^{mx} = 0 \quad \text{or} \quad e^{mx}(am^2 + bm + c) = 0.$$

Since e^{mx} is never zero for real values of x , it is apparent that the only way that this exponential function can satisfy the differential equation (2) is to choose m as a root of the quadratic equation

$$am^2 + bm + c = 0. \quad (3)$$

This last equation is called the **auxiliary equation** of the differential equation (2). Since the two roots of (3) are $m_1 = (-b + \sqrt{b^2 - 4ac})/2a$ and $m_2 = (-b - \sqrt{b^2 - 4ac})/2a$, there will be three forms of the general solution of (1) corresponding to the three cases:

- m_1 and m_2 are real and distinct ($b^2 - 4ac > 0$),
- m_1 and m_2 are real and equal ($b^2 - 4ac = 0$), and
- m_1 and m_2 are conjugate complex numbers ($b^2 - 4ac < 0$).

We discuss each of these cases in turn.

Case I: Distinct Real Roots Under the assumption that the auxiliary equation (3) has two unequal real roots m_1 and m_2 , we find two solutions, $y_1 = e^{m_1 x}$ and $y_2 = e^{m_2 x}$, respectively. We see that these functions are linearly independent on $(-\infty, \infty)$ and hence form a fundamental set. It follows that the general solution of (2) on this interval is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}. \quad (4)$$

Case II: Repeated Real Roots When $m_1 = m_2$ we necessarily obtain only one exponential solution, $y_1 = e^{m_1 x}$. From the quadratic formula we find that $m_1 = -b/2a$ since the only way to have $m_1 = m_2$ is to have $b^2 - 4ac = 0$. It follows from the discussion in Section 3.2 that a second solution of the equation is

$$y_2 = e^{m_1 x} \int \frac{e^{2m_1 x}}{e^{2m_1 x}} dx = e^{m_1 x} \int dx = x e^{m_1 x}. \quad (5)$$

In (5) we have used the fact that $-b/a = 2m_1$. The general solution is then

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}. \quad (6)$$

Case III: Conjugate Complex Roots If m_1 and m_2 are complex, then we can write $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Formally, there is no difference between this case and Case I, hence

$$y = C_1 e^{(\alpha+i\beta)x} + C_2 e^{(\alpha-i\beta)x}.$$

However, in practice we prefer to work with real functions instead of complex exponentials. To this end we use **Euler's formula**:

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

where θ is any real number.* It follows from this formula that

$$e^{i\beta x} = \cos \beta x + i \sin \beta x \quad \text{and} \quad e^{-i\beta x} = \cos \beta x - i \sin \beta x, \quad (7)$$

where we have used $\cos(-\beta x) = \cos \beta x$ and $\sin(-\beta x) = -\sin \beta x$. Note that by first adding and then subtracting the two equations in (7), we obtain, respectively,

$$e^{i\beta x} + e^{-i\beta x} = 2 \cos \beta x \quad \text{and} \quad e^{i\beta x} - e^{-i\beta x} = 2i \sin \beta x.$$

Since $y = C_1 e^{(\alpha+i\beta)x} + C_2 e^{(\alpha-i\beta)x}$ is a solution of (2) for any choice of the constants C_1 and C_2 , the choices $C_1 = C_2 = 1$ and $C_1 = 1, C_2 = -1$ give, in turn, two solutions:

$$y_1 = e^{(\alpha+i\beta)x} + e^{(\alpha-i\beta)x} \quad \text{and} \quad y_2 = e^{(\alpha+i\beta)x} - e^{(\alpha-i\beta)x}.$$

But $y_1 = e^{\alpha x}(e^{i\beta x} + e^{-i\beta x}) = 2e^{\alpha x} \cos \beta x$

and $y_2 = e^{\alpha x}(e^{i\beta x} - e^{-i\beta x}) = 2ie^{\alpha x} \sin \beta x.$

*A formal derivation of Euler's formula can be obtained from the Maclaurin series $e^x = \sum_{n=0}^{\infty} x^n/n!$ by substituting $x = i\theta$, using $i^2 = -1$, $i^3 = -i$, ..., and then separating the series into real and imaginary parts. The plausibility thus established, we can adopt $\cos \theta + i \sin \theta$ as the *definition* of $e^{i\theta}$.

Hence from Corollary (a) of Theorem 3.1.2 the last two results show that $e^{\alpha x} \cos \beta x$ and $e^{\alpha x} \sin \beta x$ are *real* solutions of (2). Moreover, these solutions form a fundamental set on $(-\infty, \infty)$. Consequently, the general solution is

$$y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x). \quad (8)$$

EXAMPLE 1 Second-Order DEs

Solve the following differential equations.

(a) $2y'' - 5y' - 3y = 0$ (b) $y'' - 10y' + 25y = 0$ (c) $y'' + 4y' + 7y = 0$

SOLUTION We give the auxiliary equations, the roots, and the corresponding general solutions.

(a) $2\lambda^2 - 5\lambda - 3 = 0$ $(2\lambda + 1)(\lambda - 3) = 0$, $\lambda_1 = -\frac{1}{2}$, $\lambda_2 = 3$. From (4),

$$y = c_1 e^{-x/2} + c_2 e^{3x}.$$

(b) $\lambda^2 - 10\lambda + 25 = 0$ $(\lambda - 5)^2 = 0$, $\lambda_1 = \lambda_2 = 5$. From (6),

$$y = c_1 e^{5x} + c_2 x e^{5x}.$$

(c) $\lambda^2 + 4\lambda + 7 = 0$, $\lambda_1 = -2 + \sqrt{3}i$, $\lambda_2 = -2 - \sqrt{3}i$. From (8) with $\alpha = -2$, $\beta = \sqrt{3}$, we have

$$y = e^{-2x} (c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x).$$

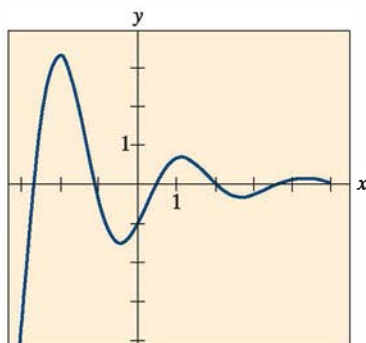


FIGURE 3.3.1 Graph of solution of IVP in Example 2

EXAMPLE 2 An Initial-Value Problem

Solve the initial-value problem $4y'' + 4y' + 17y = 0$, $y(0) = -1$, $y'(0) = 2$.

SOLUTION By the quadratic formula we find that the roots of the auxiliary equation $4\lambda^2 + 4\lambda + 17 = 0$ are $\lambda_1 = -\frac{1}{2} + 2i$ and $\lambda_2 = -\frac{1}{2} - 2i$. Thus from (8) we have $y = e^{-x/2} (c_1 \cos 2x + c_2 \sin 2x)$. Applying the condition $y(0) = -1$, we see from $e^0 (c_1 \cos 0 + c_2 \sin 0) = -1$ that $c_1 = -1$. Differentiating $y = e^{-x/2} (-\cos 2x + c_2 \sin 2x)$ and then using $y'(0) = 2$ gives $2c_2 + \frac{1}{2} = 2$ or $c_2 = \frac{3}{4}$. Hence the solution of the IVP is $y = e^{-x/2} (-\cos 2x + \frac{3}{4} \sin 2x)$. In FIGURE 3.3.1 we see that the solution is oscillatory but $y \rightarrow 0$ as $x \rightarrow \infty$.

Two Equations Worth Knowing

The two differential equations

$$y'' + k^2 y = 0 \quad \text{and} \quad y'' - k^2 y = 0,$$

k real, are important in applied mathematics. For $y'' + k^2 y = 0$, the auxiliary equation $\lambda^2 + k^2 = 0$ has imaginary roots $\lambda_1 = ki$ and $\lambda_2 = -ki$. With $\alpha = 0$ and $\beta = k$ in (8), the general solution of the DE is seen to be

$$y = c_1 \cos kx + c_2 \sin kx. \quad (9)$$

On the other hand, the auxiliary equation $\lambda^2 - k^2 = 0$ for $y'' - k^2 y = 0$ has distinct real roots $\lambda_1 = k$ and $\lambda_2 = -k$ and so by (4) the general solution of the DE is

$$y = c_1 e^{kx} + c_2 e^{-kx}. \quad (10)$$

Notice that if we choose $c_1 = c_2 = \frac{1}{2}$ and $c_1 = \frac{1}{2}$, $c_2 = -\frac{1}{2}$ in (10), we get the particular solutions $y = \frac{1}{2}(e^{kx} + e^{-kx}) = \cosh kx$ and $y = \frac{1}{2}(e^{kx} - e^{-kx}) = \sinh kx$. Since $\cosh kx$ and $\sinh kx$ are linearly independent on any interval of the x -axis, an alternative form for the general solution of $y'' - k^2 y = 0$ is

$$y = c_1 \cosh kx + c_2 \sinh kx. \quad (11)$$

See Problems 41, 42, and 62 in Exercises 3.3.

Higher-Order Equations In general, to solve an n th-order differential equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0, \quad (12)$$

where the a_i , $i = 0, 1, \dots, n$ are real constants, we must solve an n th-degree polynomial equation

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_2 \lambda^2 + a_1 \lambda + a_0 = 0. \quad (13)$$

If all the roots of (13) are real and distinct, then the general solution of (12) is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}.$$

It is somewhat harder to summarize the analogues of Cases II and III because the roots of an auxiliary equation of degree greater than two can occur in many combinations. For example, a fifth-degree equation could have five distinct real roots, or three distinct real and two complex roots, or one real and four complex roots, or five real but equal roots, or five real roots but with two of them equal, and so on. When λ_1 is a root of multiplicity k of an n th-degree auxiliary equation (that is, k roots are equal to λ_1), it can be shown that the linearly independent solutions are

$$e^{m_1 x}, x e^{m_1 x}, x^2 e^{m_1 x}, \dots, x^{k-1} e^{m_1 x}$$

and the general solution must contain the linear combination

$$c_1 e^{m_1 x} + c_2 x e^{m_1 x} + c_3 x^2 e^{m_1 x} + \cdots + c_k x^{k-1} e^{m_1 x}.$$

Lastly, it should be remembered that when the coefficients are real, complex roots of an auxiliary equation always appear in conjugate pairs. Thus, for example, a cubic polynomial equation can have at most two complex roots.

EXAMPLE 3 Third-Order DE

Solve $y''' + 3y'' - 4y = 0$.

SOLUTION It should be apparent from inspection of $\lambda^3 + 3\lambda^2 - 4 = 0$ that one root is $\lambda_1 = 1$ and so $\lambda - 1$ is a factor of $\lambda^3 + 3\lambda^2 - 4$. By division we find

$$\lambda^3 + 3\lambda^2 - 4 = (\lambda - 1)(\lambda^2 + 4\lambda + 4) = (\lambda - 1)(\lambda + 2)^2,$$

and so the other roots are $\lambda_2 = \lambda_3 = -2$. Thus the general solution is

$$y = c_1 e^x + c_2 e^{-2x} + c_3 x e^{-2x}.$$

EXAMPLE 4 Fourth-Order DE

Solve $\frac{d^4 y}{dx^4} + 2 \frac{d^2 y}{dx^2} + y = 0$.

SOLUTION The auxiliary equation $\lambda^4 + 2\lambda^2 + 1 = (\lambda^2 + 1)^2 = 0$ has roots $\lambda_1 = \lambda_3 = i$ and $\lambda_2 = \lambda_4 = -i$. Thus from Case II the solution is

$$y = C_1 e^{ix} + C_2 e^{-ix} + C_3 x e^{ix} + C_4 x e^{-ix}.$$

By Euler's formula the grouping $C_1 e^{ix} + C_2 e^{-ix}$ can be rewritten as $c_1 \cos x + c_2 \sin x$ after a relabeling of constants. Similarly, $x(C_3 e^{ix} + C_4 e^{-ix})$ can be expressed as $x(c_3 \cos x + c_4 \sin x)$. Hence the general solution is

$$y = c_1 \cos x + c_2 \sin x + c_3 x \cos x + c_4 x \sin x.$$

Example 4 illustrates a special case when the auxiliary equation has repeated complex roots. In general, if $r_1 = \alpha + i\beta$, $\beta > 0$, is a complex root of multiplicity k of an auxiliary equation with real coefficients, then its conjugate $r_2 = \alpha - i\beta$ is also a root of multiplicity k . From the $2k$ complex-valued solutions

$$\begin{aligned} e^{(\alpha+i\beta)x}, & xe^{(\alpha+i\beta)x}, x^2e^{(\alpha+i\beta)x}, \dots, x^{k-1}e^{(\alpha+i\beta)x} \\ e^{(\alpha-i\beta)x}, & xe^{(\alpha-i\beta)x}, x^2e^{(\alpha-i\beta)x}, \dots, x^{k-1}e^{(\alpha-i\beta)x} \end{aligned}$$

we conclude, with the aid of Euler's formula, that the general solution of the corresponding differential equation must then contain a linear combination of the $2k$ real linearly independent solutions

$$\begin{aligned} e^{\alpha x} \cos \beta x, & xe^{\alpha x} \cos \beta x, x^2e^{\alpha x} \cos \beta x, \dots, x^{k-1}e^{\alpha x} \cos \beta x \\ e^{\alpha x} \sin \beta x, & xe^{\alpha x} \sin \beta x, x^2e^{\alpha x} \sin \beta x, \dots, x^{k-1}e^{\alpha x} \sin \beta x. \end{aligned}$$

In Example 4 we identify $k = 2$, $\alpha = 0$, and $\beta = 1$.

Rational Roots Of course the most difficult aspect of solving constant-coefficient differential equations is finding roots of auxiliary equations of degree greater than two. For example, to solve $3y''' + 5y'' + 10y' - 4y = 0$ we must solve $3r^3 + 5r^2 + 10r - 4 = 0$. Something we can try is to test the auxiliary equation for rational roots. Recall, if $r = p/q$ is a rational root (expressed in lowest terms) of an auxiliary equation $a_n r^n + \dots + a_1 r + a_0 = 0$ with integer coefficients, then p is a factor of a_0 and q is a factor of a_n . For our specific cubic auxiliary equation, all the factors of $a_0 = -4$ and $a_n = 3$ are $p: \pm 1, \pm 2, \pm 4$ and $q: \pm 1, \pm 3$, so the possible rational roots are $p/q: \pm 1, \pm 2, \pm 4, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{4}{3}$. Each of these numbers can then be tested, say, by synthetic division. In this way we discover both the root $r_1 = \frac{1}{3}$ and the factorization

$$3r^3 + 5r^2 + 10r - 4 = (r - \frac{1}{3})(3r^2 + 6r + 12).$$

The quadratic formula then yields the remaining roots $r_2 = -1 + \sqrt{3}i$ and $r_3 = -1 - \sqrt{3}i$. Therefore the general solution of $3y''' + 5y'' + 10y' - 4y = 0$ is $y = c_1 e^{x/3} + e^{-x}(c_2 \cos \sqrt{3}x + c_3 \sin \sqrt{3}x)$.

Use of Computers Finding roots or approximations of roots of polynomial equations is a routine problem with an appropriate calculator or computer software. The computer algebra systems *Mathematica* and *Maple* can solve polynomial equations (in one variable) of degree less than five in terms of algebraic formulas. For the auxiliary equation in the preceding paragraph, the commands

`Solve[3 m^3 + 5 m^2 + 10 m - 4 == 0, m]` (in *Mathematica*)

`solve(3*m^3 + 5*m^2 + 10*m - 4, m);` (in *Maple*)

yield immediately their representations of the roots $\frac{1}{3}, -1 + \sqrt{3}i, -1 - \sqrt{3}i$. For auxiliary equations of higher degree it may be necessary to resort to numerical commands such as **NSolve** and **FindRoot** in *Mathematica*. Because of their capability of solving polynomial equations, it is not surprising that some computer algebra systems are also able to give explicit solutions of homogeneous linear constant-coefficient differential equations. For example, the inputs

`DSolve [y''[x] + 2 y'[x] + 2 y[x] == 0, y[x], x]` (in *Mathematica*)

`dsolve(diff(y(x), x$2) + 2*diff(y(x), x) + 2*y(x) = 0, y(x));` (in *Maple*)

give, respectively,

$$y[x] \rightarrow \frac{C[2] \cos [x] - C[1] \sin [x]}{E^x} \quad (14)$$

and

$$y(x) = _C1 \exp(-x) \sin(x) - _C2 \exp(-x) \cos(x)$$

Translated, this means $y = c_2 e^{-x} \cos x + c_1 e^{-x} \sin x$ is a solution of $y'' + 2y' + 2y = 0$.

In the classic text *Differential Equations* by Ralph Palmer Agnew* (used by the author as a student), the following statement is made:

It is not reasonable to expect students in this course to have computing skills and equipment necessary for efficient solving of equations such as

$$4.317 \frac{d^4 y}{dx^4} + 2.179 \frac{d^3 y}{dx^3} + 1.416 \frac{d^2 y}{dx^2} + 1.295 \frac{dy}{dx} + 3.169y = 0. \quad (15)$$

Although it is debatable whether computing skills have improved in the intervening years, it is a certainty that technology has. If one has access to a computer algebra system, equation (15) could be considered reasonable. After simplification and some relabeling of the output, *Mathematica* yields the (approximate) general solution

$$y = c_1 e^{-0.728852x} \cos(0.618605x) + c_2 e^{-0.728852x} \sin(0.618605x) \\ + c_3 e^{-0.476478x} \cos(0.759081x) + c_4 e^{-0.476478x} \sin(0.759081x).$$

We note in passing that the **DSolve** and **dsolve** commands in *Mathematica* and *Maple*, like most aspects of any CAS, have their limitations.

Finally, if we are faced with an initial-value problem consisting of, say, a fourth-order differential equation, then to fit the general solution of the DE to the four initial conditions we must solve a system of four linear equations in four unknowns (the c_1, c_2, c_3, c_4 in the general solution). Using a CAS to solve the system can save lots of time. See Problems 35, 36, 69, and 70 in Exercises 3.3.

*McGraw-Hill, New York, 1960.

Remarks

In case you are wondering, the method of this section also works for homogeneous linear *first-order* differential equations $ay' + by = 0$ with constant coefficients. For example, to solve, say, $2y' + 7y = 0$, we substitute $y = e^{mx}$ into the DE to obtain the auxiliary equation $2m + 7 = 0$. Using $m = -\frac{7}{2}$, the general solution of the DE is then $y = c_1 e^{-7x/2}$.

3.3 Exercises Answers to selected odd-numbered problems begin on page ANS-5.

In Problems 1–14, find the general solution of the given second-order differential equation.

1. $4y'' + y' = 0$
2. $y'' - 36y = 0$
3. $y'' - y' - 6y = 0$
4. $y'' - 3y' + 2y = 0$
5. $y'' + 8y' + 16y = 0$
6. $y'' - 10y' + 25y = 0$
7. $12y'' - 5y' - 2y = 0$
8. $y'' + 4y' - y = 0$
9. $y'' + 9y = 0$
10. $3y'' + y = 0$
11. $y'' - 4y' + 5y = 0$
12. $2y'' + 2y' + y = 0$
13. $3y'' + 2y' + y = 0$
14. $2y'' - 3y' + 4y = 0$

In Problems 15–28, find the general solution of the given higher-order differential equation.

15. $y''' - 4y'' - 5y' = 0$

16. $y''' - y = 0$

17. $y''' - 5y'' + 3y' + 9y = 0$

18. $y''' + 3y'' - 4y' - 12y = 0$

19. $\frac{d^3 u}{dt^3} + \frac{d^2 u}{dt^2} - 2u = 0$

20. $\frac{d^3 x}{dt^3} - \frac{d^2 x}{dt^2} - 4x = 0$

21. $y''' + 3y'' + 3y' + y = 0$

22. $y''' - 6y'' + 12y' - 8y = 0$

23. $y^{(4)} + y''' + y'' = 0$

24. $y^{(4)} - 2y'' + y = 0$